

Move Away from Single Server

As the number of users and uploaded files increases, the limitations of the single server architecture quickly become visible.

Eventually, the storage server becomes full and users can no longer upload files.

Storage Capacity Problem

The server starts showing low storage warnings as illustrated in **Figure**.



Figure demonstrates a critical situation where only **10 MB** of storage space remains available.

At this point:

- File uploads fail
- User experience degrades
- The platform becomes unreliable

This is a serious scalability problem.

Sharding the Storage Servers

To solve the storage limitation issue, the first improvement is to distribute files across multiple storage servers.

A common approach is:

- Partition users based on `user_id`
- Store different user files on different storage nodes

This process is known as **sharding**.

Storage Sharding by User ID

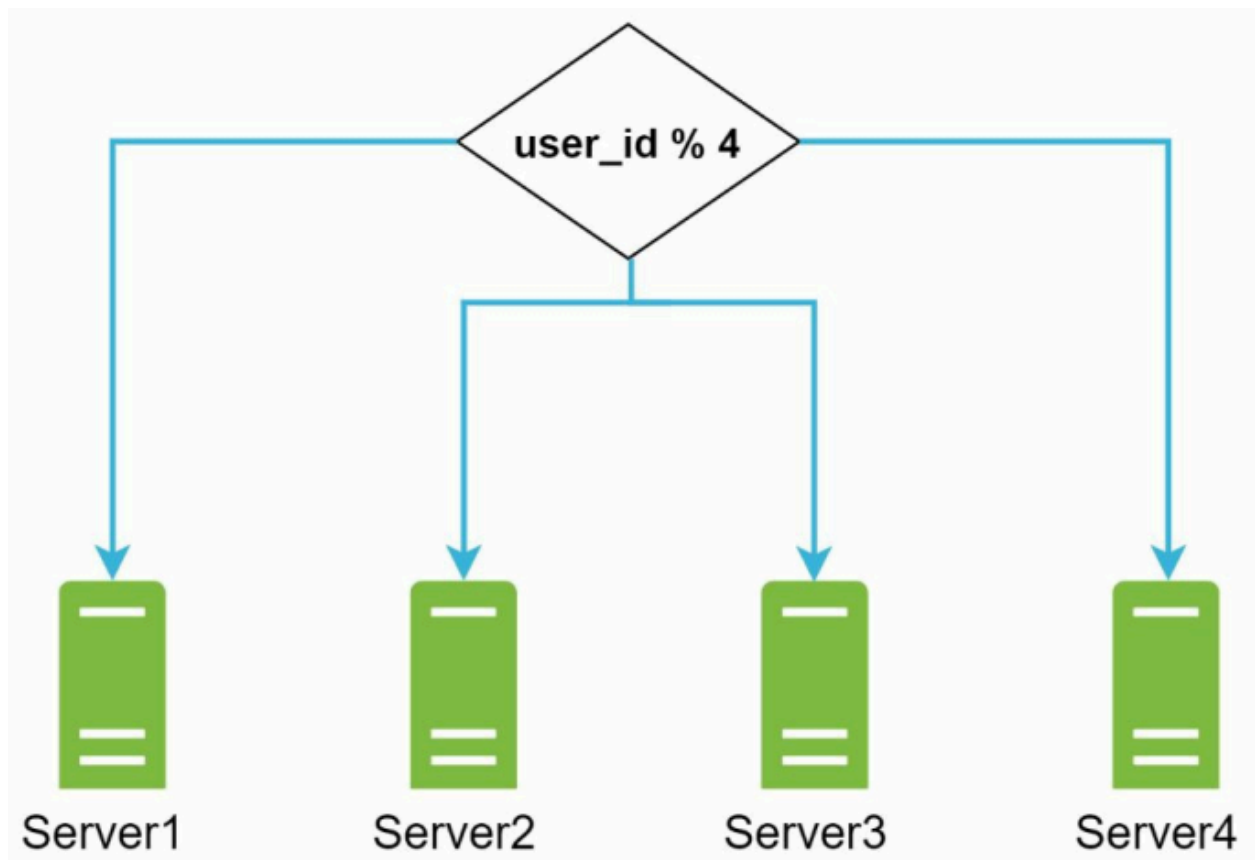


Figure illustrates how files are distributed across multiple storage servers using `user_id`-based sharding.

Example:

User ID Range	Storage Server
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1 – 1M	Storage Server A
1M – 2M	Storage Server B
2M – 3M	Storage Server C

Benefits of sharding:

- Increased storage capacity
- Better scalability
- Reduced load per server
- Parallel processing capability

However, sharding alone does not solve reliability issues.

Reliability Concern

Even after storage sharding, another major issue remains:

What happens if a storage server crashes?

Possible consequences:

- Permanent file loss
- Service downtime
- Corrupted uploads
- Unavailable user files

To prevent data loss, the system requires replication and durable distributed storage.

Migrating to Object Storage

After researching industry practices, the architecture adopts object storage such as:

- [Amazon S3](#)

Amazon S3 provides:

- High durability
- High scalability
- Built-in replication
- Strong availability guarantees
- Object-based storage model

S3 is widely used by large-scale companies such as:

- Netflix
- Airbnb
- Dropbox-like systems

What is Amazon S3?

Amazon S3 (Simple Storage Service) is an object storage system where:

- Files are stored as objects
- Each object has a unique key
- Objects are stored inside buckets

A bucket is conceptually similar to a folder or namespace.

Same-Region and Cross-Region Replication

Amazon S3 supports automatic replication across multiple regions.

This greatly improves:

- Durability
- Availability
- Disaster recovery

S3 Replication Architecture

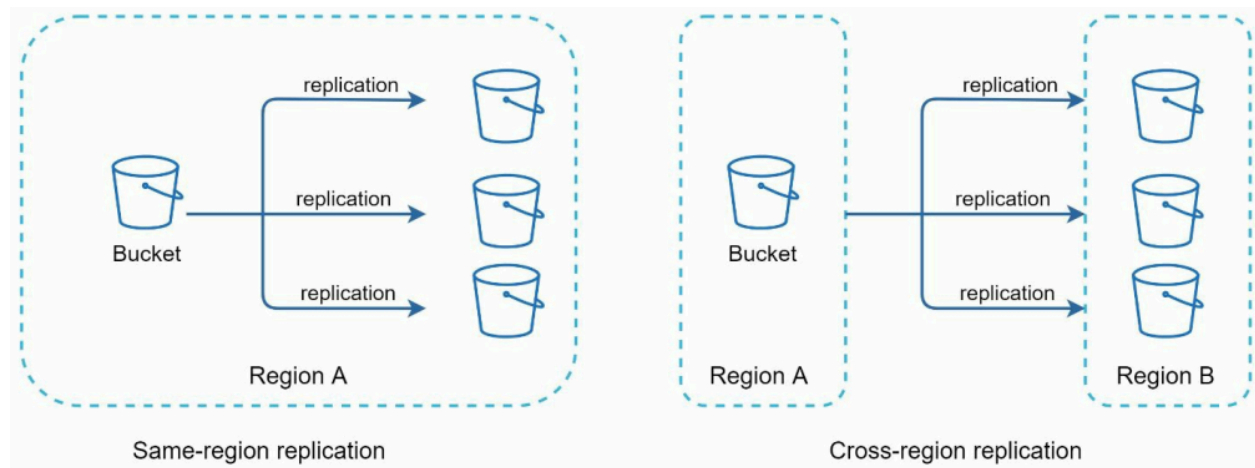


Figure shows two replication approaches.

Same-Region Replication (Left Side)

Data is replicated across multiple data centers within the same geographic region.

Benefits:

- Fast replication
- Protection against local hardware failures

Cross-Region Replication (Right Side)

Data is replicated across geographically separated regions.

Benefits:

- Disaster recovery
- Regional outage protection
- High availability

Example:

- Original file stored in US-East
- Replica stored in Europe or Asia

This ensures files remain accessible even if an entire region becomes unavailable.

Advantages of Using S3

Using distributed object storage solves multiple problems simultaneously.

Durability

Files are replicated automatically across multiple storage devices and regions.

Scalability

Storage can grow from terabytes to petabytes seamlessly.

High Availability

Users can still access files during server or regional failures.

Operational Simplicity

The engineering team no longer needs to:

- Manage disks manually
- Handle replication logic
- Replace failed storage nodes

Additional System Improvements

After solving the storage problem, several additional architectural improvements are introduced.

1. Load Balancer

A load balancer is added in front of the web servers.

Responsibilities:

- Distribute incoming traffic evenly
- Prevent server overload
- Detect failed servers
- Redirect traffic automatically

Benefits:

- Better fault tolerance
- Improved scalability
- High availability

2. Multiple Web Servers

With the load balancer in place, web servers can now scale horizontally.

Benefits:

- Add servers during traffic spikes
- Remove servers during low traffic
- Support millions of concurrent users

Web servers become stateless so they can be replaced easily.

3. Distributed Metadata Database

The metadata database is moved out of the single server setup.

Enhancements include:

- Database replication
- Database sharding
- Separate metadata service

This avoids:

- Single point of failure
- Database bottlenecks

The metadata database stores:

- User information
- File metadata
- Sharing permissions
- Revision history
- Sync state

4. Distributed File Storage

Instead of local disk storage:

- Files are stored in Amazon S3
- Replicated across multiple regions

Benefits:

- Extremely high durability
- Disaster recovery
- Global availability

Updated High-Level Architecture

After applying all optimizations, the architecture becomes fully distributed.

The system now contains:

- Load balancer
- Multiple stateless web/API servers
- Distributed metadata database
- Amazon S3 object storage
- Replication across regions

This removes tight coupling between components.

Distributed Google Drive Architecture

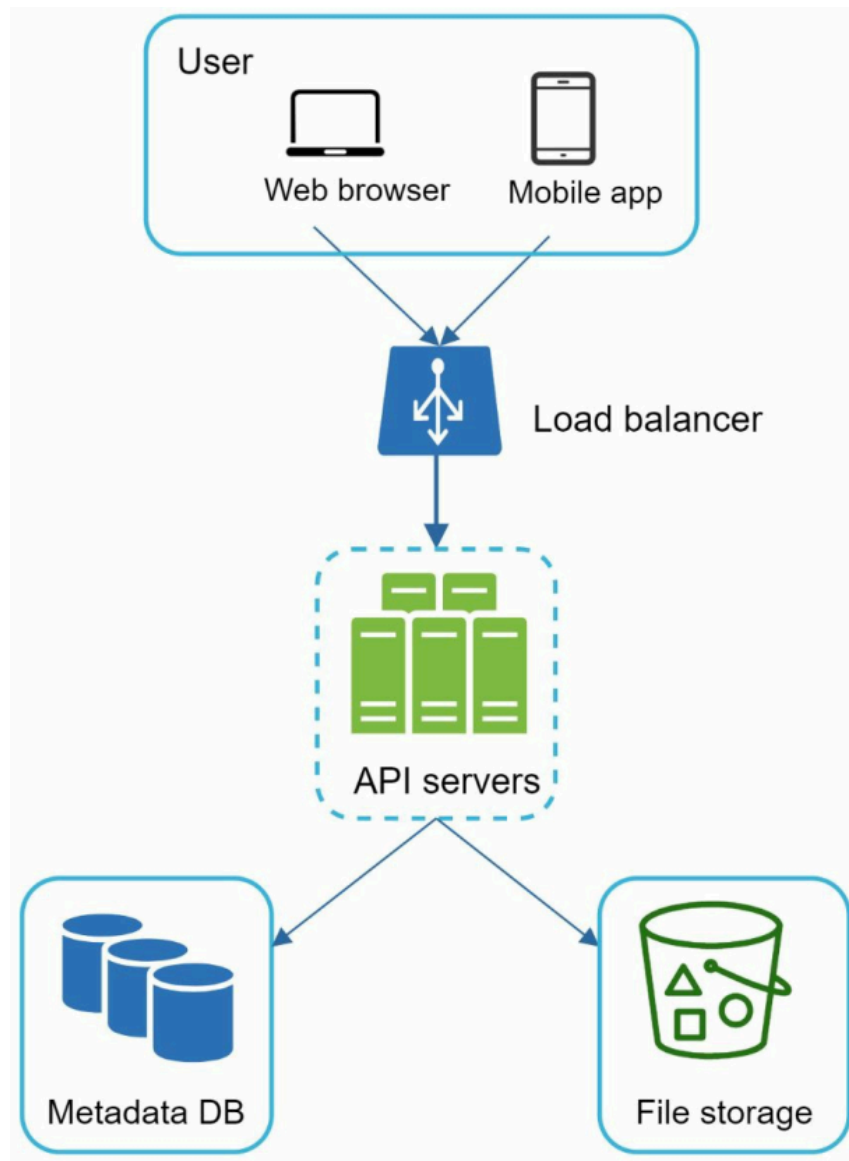


Figure presents the updated distributed architecture after decoupling the major components.

The architecture now includes:

- Load balancer
- Scalable web servers
- Replicated metadata database
- Distributed S3 storage

This design provides:

- High scalability
- High availability
- Reliability
- Better fault tolerance

Key Architectural Improvements

Problem	Solution
Limited storage	Storage sharding
Data loss risk	S3 replication
Single server bottleneck	Load balancer + multiple servers
Database SPOF	Replication + sharding
Poor scalability	Distributed architecture

Result

The system evolves from:

- A simple monolithic single server

To:

- A highly scalable distributed cloud storage platform

This architecture can now support:

- Millions of users
- Petabyte-scale storage
- Reliable synchronization
- Global availability
- High durability and fault tolerance